

The exploration cost of learning boundary and population causal effects

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Abstract

Many deployed systems assign actions with a known policy that becomes more decisive over time, for example a softmax over reward-model scores with a falling temperature. We ask which causal effects the resulting logs can deliver and what it costs to learn them. Cost is measured by exploration loss, the value the deployer gives up, in its own objective, each time the logged action is not the preferred one. In a model with a one-dimensional score, Gaussian outcomes and Lipschitz mean functions, a single shrinking temperature yields Wald intervals for the effect at the decision boundary that are asymptotically valid uniformly over the model, and the exploration loss goes to zero. In the same model, uniformly consistent estimation of the population average effect requires the worst-case loss to diverge, whatever the design and estimator. This contrast needs an operational cost that vanishes at the boundary. When the mean functions are only bounded, a given precision for the population effect needs loss of order the squared mean root gap divided by the squared error, however large the sample. A common temperature needs loss growing almost linearly in the sample size; uniform mixing keeps it bounded.

Keywords: adaptive experiments, limited overlap, regression discontinuity, minimax lower bounds, generative AI, off-policy evaluation

MSC 2020: 62D20, 62K05, 62C20, 62G20

1 Introduction

Many systems in use today choose actions with a policy that grows more decisive as it gains confidence. Bandit algorithms work this way, as do ranking systems, algorithmic eligibility rules and conversational AI systems that pick a response strategy through a softmax over reward-model scores. Their logs record the action and usually the probability with which it was taken. This paper asks what such logs can tell us about causal effects when the policy is known and close to deterministic, and what it costs to learn more.

Political persuasion by language models is a case in point. Conversations with these models move political attitudes [1, 2], and the strategy a model uses, say presenting evidence or asking a question, is itself chosen adaptively. Researchers and regulators who want to know what these strategies do will often have only the deployment logs to work with.

Two targets have to be kept apart. The *boundary effect* is the effect among contexts where the policy is indifferent between the two actions. It describes units at the margin of the current policy and governs small shifts in the policy’s threshold, much like the marginal policy-relevant treatment effect of Carneiro, Heckman and Vytlacil [3]. The *population average effect* is the effect of assigning one action instead of the other across every context the system serves. Knowing one well says little about the other.

We measure the cost of learning in the deployer’s own terms. Whenever the logged action differs from the one its objective prefers, the deployer loses an operational value gap $g(h)$, and we call the running total the *exploration loss* R_n . For a system that acts on a score, the obvious gap is the score difference, which is small where the policy is nearly indifferent and large where it is confident. Since g comes from the deployer’s objective rather than from the outcome under study, it is known when the design is chosen.

With a single temperature that shrinks at a suitable rate, the boundary effect can be estimated with valid confidence intervals while the exploration loss goes to zero. In the same model, no design and no estimator learn the population effect uniformly unless the worst-case loss diverges. The ingredients are familiar: limited overlap, local randomization at a threshold and cost-weighted allocation have all been studied. What is new is that both targets are treated in one model and on one cost scale, with lower bounds that hold for every estimator and every adaptive design. The contrast depends on a cost that vanishes at the boundary. If each deviation costs a fixed amount, the cumulative exploration loss of boundary inference under a common temperature also diverges (Proposition 1).

Figure 1 shows the construction behind the lower bound. The two outcome models agree near the boundary and differ only for the unpreferred action further away, which a sharpening policy almost never takes.

The paper makes three contributions. Theorems 1 and 3 establish the separation in a model with Gaussian outcomes and globally Lipschitz mean functions. With $\tau_n = n^{-\zeta}$ and $\zeta \in (1/2, 1)$, Wald intervals for the boundary effect are uniformly valid and $R_n \rightarrow 0$, while uniformly consistent estimation of the population effect requires $\sup R_n \rightarrow \infty$. Theorem 2 is an information–exploration inequality for all estimators and adaptive designs when the mean functions are only bounded, and Proposition 2 shows that a gap-based design attains its order. Section 7 turns to design. A single common temperature is costly for the population effect; uniform mixing, context-specific temperatures and some temperature mixtures are not.

Throughout, each unit receives one decision, contexts are independent and the assignment probabilities are known. The score and the preferred action are fixed functions of the context. By “adaptive” we mean sharpening, or designs that use past data, and not a bandit that learns from outcomes which action is better. Other limitations are collected in Section 9.

The paper is organized as follows. Section 2 sets out the model and Section 3 previews

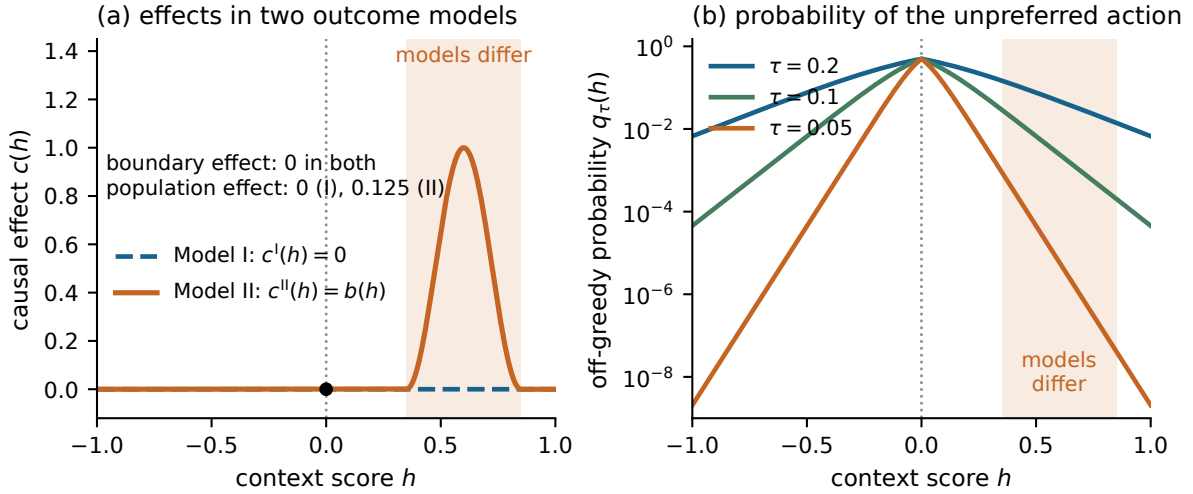


Figure 1: Construction behind Theorem 3 (Model I is μ^0 , Model II is μ^t), with $H \sim U(-1, 1)$, $g(h) = |h|$, $u_1 = 0.35$, $u_2 = 0.85$ and $t = 1$. Left: effect functions. Both boundary effects are zero; the population effects are 0 and 0.125. Right: probability of the unpreferred action at three temperatures. The models differ only in the shaded region.

the results. Sections 4, 5 and 6 treat the boundary effect, the population effect and the separation. Section 7 discusses temperature as a design choice, Section 8 reports simulations and Section 9 concludes. Proofs are in the appendix.

Related work. The boundary result is closest to work that uses algorithmic or mechanism-based assignment as a source of local randomization. Lee [4] reads regression discontinuity as local randomization. Abdulkadiroğlu et al. [5, 6] use propensity scores induced by school-assignment mechanisms, including local scores near tie-breaking cutoffs, and Narita and Yata [7] show that overlap-type weights built from an approximate propensity score of a known algorithm converge to a boundary estimand. In their work the analyst chooses how much to smooth. Here the system’s temperature plays that role, and we keep track of the exploration loss that comes with it. Eckles et al. [8] exploit known assignment noise of fixed scale at a cutoff. Overlap weights go back to Li, Morgan and Zaslavsky [9], and Humphreys [10] shows that fixed-effects regression weights stratum effects by $p_j(1 - p_j)$; under a sharpening policy almost all of that weight sits at the boundary.

The trade-off between reward and estimation has mostly been studied for bandits without contexts [11, 12], where the guarantees concern particular algorithms. Liang and Bojinov [13] are closest to our design conclusions. They mix any bandit algorithm with a Bernoulli design whose weight decays more slowly than $t^{-1/4}$ and obtain anytime-valid confidence sequences for the average effect. Their setting has no contexts and no cost for the Bernoulli assignments; we charge each departure from the deployer’s preferred action by its value gap, and Theorem efthm:info shows that some mixing of this kind cannot be avoided by any design that learns the population effect uniformly. Hadad et

al. [14] build confidence intervals for policy values from bandit data when assignment probabilities decay at a polynomial rate. Under a sharpening policy the probabilities decay exponentially in $|\Delta|/\tau$ away from the boundary, and our boundary estimator targets β_0 rather than a policy value. Adaptive designs have also been proposed for efficient estimation of average effects [15], for policy choice [16] and for applications in political science [17]; in those designs assignment serves estimation or choice, whereas here it is fixed by the deployer’s objective. Conservative bandits [18] keep cumulative reward above that of a baseline policy, and Wan, Kveton and Song [19] optimize exploration for policy evaluation under a value constraint; both measure cost against a baseline, as we do, but their guarantees concern the regret or evaluation error of the proposed algorithms. Tucker and Joachims [20] and Douglas, Persson and Provost [21] choose logging policies to reduce the variance of off-policy estimates, and Koppel et al. [22] balance regret against the quality of average-effect estimates in contextual bandits. The allocation $q \propto g^{-1/2}$ is optimum stratified allocation with costs [23] and appears for factorial experiments in [24]; what we add is that its constant bounds every estimator and adaptive design. Li and Imai [25] compare ways of evaluating an individualized rule without charging for departures from it, and Cesa-Bianchi et al. [26] show that Boltzmann exploration with one schedule can fail for regret. Theorem 4 is an analogue for estimation.

The lower bounds draw on off-policy evaluation [27–29] and on limited overlap and trimming [30–34], in particular the irregular identification of Khan and Tamer [30]. Ring and Schomaker [35] use positivity diagnostics to choose alternative estimands, and Khan, Saveski and Ugander [36] bound effects without overlap by assuming smoothness, the kind of extrapolation our lower bounds rule out. For generative treatments, Athey, Imbens and Ji [37] identify effects of content features from stochastic algorithms, Fresh and Shin [38] classify conversational estimands, and Imai and Nakamura [39, 40] study effects of realized text. We stay at the level of strategy assignment and hold the generator fixed.

2 Setup

Treatments. Units $i = 1, \dots, n$ arrive with contexts $H_i \sim F$. The system chooses a strategy $A_i \in \{0, 1\}$ and produces a message $M_i \sim G(\cdot \mid A_i, H_i)$ from a fixed generation kernel G , which includes token-level sampling settings; the outcome is Y_i . The treatment is the assignment of the strategy with G held fixed, so potential outcomes $Y_i(a)$ include the randomness of the message [41, 42]; changing G changes the treatment. We assume:

- (C1) *consistency*: $Y_i = Y_i(A_i)$;
- (C2) *independent units*: $(H_i, Y_i(0), Y_i(1))$ are independent and identically distributed across i , with no interference;
- (C3) *sequential ignorability with known probabilities*: A_i is independent of $(Y_i(0), Y_i(1))$ given $(H_i, \mathcal{F}_{i-1})$, where $\mathcal{F}_{i-1}$ is the data of units $1, \dots, i-1$, and $\mathbb{P}(A_i = 1 \mid H_i, \mathcal{F}_{i-1})$ is known.

Then the law of Y_i given $(A_i = a, H_i = h, \mathcal{F}_{i-1})$ is that of $Y(a)$ given $H = h$, so statements about conditional outcome laws are statements about potential outcomes.

Deployer. The deployer has a score gap $\Delta(h) = s_1(h) - s_0(h)$ from its own objective, a preferred (greedy) action $a^*(h) = \mathbf{1}\{\Delta(h) > 0\}$, and an operational value gap $g(h) \geq 0$, the value its objective loses by choosing $\bar{a}(h) = 1 - a^*(h)$. We write $g = \phi(|\Delta|)$ when the gap is a function of the score. The score and the preferred action are fixed functions of the context.

Running example. A conversational system chooses between presenting evidence ($A = 1$) and asking a question ($A = 0$). A reward model scores both strategies given the conversation so far, H ; $\Delta(H)$ is the score difference, and the system logs the probability with which it chose evidence. The operational gap $g(H)$ is the loss in the deployer’s objective, for example predicted engagement, from using the lower-scoring strategy. The outcome Y is the person’s attitude after the conversation, measured separately and not used by the policy.

Probabilities and designs. We write

$$e(h) = \mathbb{P}(A = 1 \mid H = h), \quad q(h) = \mathbb{P}(A \neq a^*(h) \mid H = h)$$

for the probability of action 1 and of the non-preferred action. A design is non-adaptive if these probabilities do not depend on past data. A *common temperature* design uses one deterministic $\tau = \tau_n$ and

$$e_\tau(h) = \text{sig}(\Delta(h)/\tau), \quad q_\tau(h) = \text{sig}(-|\Delta(h)|/\tau), \quad \text{sig}(x) = 1/(1 + \exp(-x)).$$

Exploration loss. The expected cumulative operational loss is

$$R_n = \mathbb{E} \left[\sum_{i=1}^n g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\} \right], \quad (1)$$

and $R_n = n\mathbb{E}[g(H)q(H)]$ for non-adaptive designs. Under adaptive designs R_n may depend on the outcome law through past observations.

Estimands. Let $\mu_a(h) = \mathbb{E}[Y(a) \mid H = h]$ and $c = \mu_1 - \mu_0$. The population average effect is $\theta = \mathbb{E}[c(H)]$. With a one-dimensional score $\Delta(h) = h$ the boundary effect is $\beta_0 = c(0)$. For a given temperature, the overlap-weighted effect $\beta_{\text{ov},\tau} = \mathbb{E}[\omega_\tau(H)c(H)]/\mathbb{E}[\omega_\tau(H)]$, with $\omega_\tau = e_\tau(1 - e_\tau)$ [9], is an intermediate target.

3 The results in brief

Take $\Delta(h) = h$, Gaussian outcomes with variance σ^2 , and mean functions bounded by B and L -Lipschitz; call this model $\mathcal{M}_{\text{Lip}}$. Suppose the logs come from one common temperature $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$ and that the operational gap vanishes linearly at the boundary. Then:

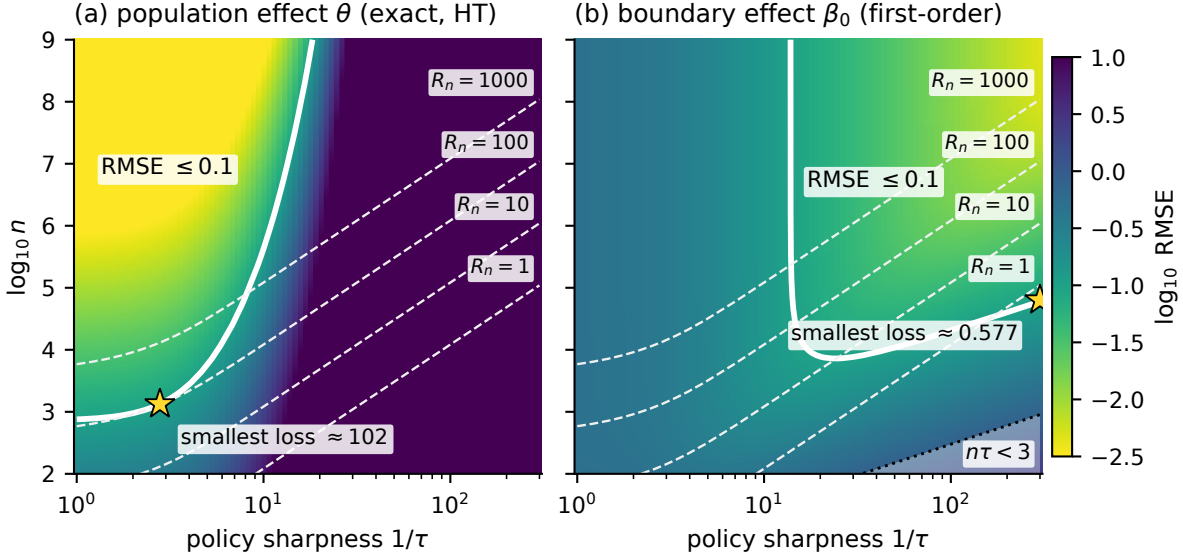


Figure 2: Root mean squared error ($\log_{10}$ scale) under one common temperature, in the design of Section 8. (a) Exact value for the Horvitz–Thompson estimator of θ . (b) First-order approximation $\{(\beta_{\text{ov},\tau} - \beta_0)^2 + s_n^2/n\}^{1/2}$ for the estimator of Theorem 1, checked by simulation in Section 8. White curves enclose error at most 0.1, dashed curves are levels of R_n , and stars mark the smallest loss reaching that precision on the grid. The approximation was not checked in the shaded region $n\tau < 3$.

- (i) the Wald interval for the boundary effect is asymptotically valid uniformly over $\mathcal{M}_{\text{Lip}}$, and $R_n \rightarrow 0$ (Theorems 1 and 3);
- (ii) $\mathcal{M}_{\text{Lip}}$ contains two outcome laws with the same boundary effect and different population effects that differ only for the non-preferred action, in a region where the gap is bounded below. Once $R_n \rightarrow 0$, no estimator is consistent for the population effect at both, and uniformly consistent estimation over $\mathcal{M}_{\text{Lip}}$ requires divergent worst-case loss (Theorem 3);
- (iii) if the mean functions are only bounded, precision δ for the population effect requires worst-case loss of order at least $(\mathbb{E}\sqrt{g})^2/\delta^2$ as $\delta \rightarrow 0$, whatever the value of n (Theorem 2).

Figure 2 shows what this looks like in the simulation design of Section 8. Under one common temperature, the Horvitz–Thompson estimator of the population effect reaches root mean squared error 0.1 only where the loss is at least about 102 on the plotted grid. The boundary effect reaches the same precision with loss 0.58 at the edge of the grid, and with less at larger n . Panel (a) describes a single estimator; the lower bounds for all estimators are Theorems 2 and 3.

4 Boundary effects at vanishing exploration loss

Assumption 1 (Boundary model). $\Delta(h) = h$, $a^*(h) = \mathbf{1}\{h > 0\}$, and

- (D1) H has a density $f \leq \bar{f}$, continuous at 0, with $f(0) > 0$;
- (D2) $|\mu_a| \leq B$, and μ_a is L -Lipschitz on $[-u_0, u_0]$;
- (D3) $\sigma_a^2(h) = \text{Var}(Y \mid A = a, H = h)$ is continuous at 0 with $\sigma_1^2(0) + \sigma_0^2(0) > 0$, and $\mathbb{E}[Y^4 \mid A, H] \leq M_4$;
- (D4) $\mathbb{P}(A = 1 \mid H = h) = \text{sig}(h/\tau_n)$ with $\tau_n \rightarrow 0$, known;
- (D5) $g = \phi(|h|) \leq \bar{G}$ with $\phi(u) \leq \bar{\kappa}u$ on $[0, u_0]$; in the strong form $\phi'(0) = \kappa > 0$.

With $e_i = e_{\tau_n}(H_i)$, define

$$\begin{aligned} w_{1i} &= A_i(1 - e_i), & w_{0i} &= (1 - A_i)e_i, \\ \hat{D}_a &= \frac{1}{n} \sum_{i=1}^n w_{ai}, & \hat{m}_a &= \frac{\sum_{i=1}^n w_{ai} Y_i}{\sum_{i=1}^n w_{ai}}, & \hat{\beta} &= \hat{m}_1 - \hat{m}_0, \end{aligned} \quad (2)$$

$$\hat{\psi}_i = \frac{w_{1i}(Y_i - \hat{m}_1)}{\hat{D}_1} - \frac{w_{0i}(Y_i - \hat{m}_0)}{\hat{D}_0}, \quad \hat{s}^2 = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2. \quad (3)$$

If either weight sum is zero, set $\hat{\beta} = \hat{s}^2 = 0$ and report the interval $\mathbb{R}$; this event has probability tending to zero when $n\tau_n \rightarrow \infty$. Write $m_\tau = \mathbb{E}[\omega_\tau(H)]$, $\bar{\mu}_a = \mathbb{E}[\omega_\tau(H)\mu_a(H)]/m_\tau$,

$$\xi = \frac{w_1(Y - \bar{\mu}_1) - w_0(Y - \bar{\mu}_0)}{m_\tau}, \quad s_n^2 = \text{Var}(\xi),$$

so that $\bar{\mu}_1 - \bar{\mu}_0 = \beta_{\text{ov}, \tau}$ and ξ is the linearization of $\hat{\beta} - \beta_{\text{ov}, \tau}$.

Theorem 1 (Boundary inference). *Under Assumption 1, if $n\tau_n \rightarrow \infty$:*

- (i) $(\hat{\beta} - \beta_{\text{ov}, \tau_n})/(s_n/\sqrt{n}) \Rightarrow N(0, 1)$, and $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau_n f(0)\}(1 + o(1))$;
- (ii) $|\beta_{\text{ov}, \tau_n} - \beta_0| \leq C\tau_n$ for large n , for every $C > 4 \log 2 L \bar{f}/f(0)$;
- (iii) $\hat{s}^2/s_n^2 \rightarrow 1$ in probability;
- (iv) if also $n\tau_n^3 \rightarrow 0$, the Wald interval $\hat{\beta} \pm z_{1-\eta/2} \hat{s}/\sqrt{n}$ has asymptotic coverage $1 - \eta$ for β_0 ;
- (v) $R_n = n\mathbb{E}[g(H)q_{\tau_n}(H)] \leq n\tau_n^2 \bar{\kappa} \bar{f} \pi^2/6 + n\bar{G} \exp(-u_0/\tau_n)$, and in the strong form $R_n = n\tau_n^2 \kappa f(0) \pi^2/6 (1 + o(1))$.

In particular, for $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$, the interval in (iv) is asymptotically valid and $R_n \rightarrow 0$.

The three rate conditions say that the boundary sample grows ($n\tau_n \rightarrow \infty$, $\zeta < 1$), that the bias is small next to the standard error ($n\tau_n^3 \rightarrow 0$, $\zeta > 1/3$) and that the loss vanishes ($n\tau_n^2 \rightarrow 0$, $\zeta > 1/2$). At $\zeta = 1/2$ the interval is still valid and the loss stays bounded. The error is $O_p((n\tau_n)^{-1/2} + \tau_n)$; optimality of this rate is not established here. A smoother effect helps only with more structure. If c is $C^{1,1}$ and f is Lipschitz near zero, the bias is $O(\tau^2)$ and (iv) needs only $n\tau_n^5 \rightarrow 0$, but an effect equal to $1 + |h|^{3/2}$ near zero, which is differentiable, still has bias of order $\tau^{3/2}$. Proofs are in Appendix A.

Exploration continues even as the loss vanishes. The expected number of non-preferred actions, $n\mathbb{E}[q_{\tau_n}(H)] = 2\log 2 f(0) n\tau_n(1 + o(1))$, grows with n . The loss goes to zero because these actions happen ever closer to the boundary, where they cost little. How much they cost there depends on the shape of the gap.

Proposition 1 (Shape of the operational gap). *Let (D1) and (D4) hold and $g = \phi(|h|) \leq \bar{G}$.*

- (a) *If $\phi(u) \leq \bar{\kappa}u^p$ on $[0, u_0]$ for some $p > 0$, then $R_n \leq 2\bar{\kappa}\bar{f}C_p n\tau_n^{p+1} + n\bar{G}\exp(-u_0/\tau_n)$ with $C_p = \int_0^\infty v^p \text{sig}(-v) dv < \infty$. Under the other conditions of Theorem 1, $\tau_n = n^{-\zeta}$ with $\max\{1/3, 1/(p+1)\} < \zeta < 1$ gives a valid interval for β_0 and $R_n \rightarrow 0$.*
- (b) *If $\phi(u) \geq \lambda_0 > 0$ for $u \in (0, u_0]$, then $R_n \geq 2\log 2 \lambda_0 f(0) n\tau_n(1 + o(1))$, which diverges whenever $n\tau_n \rightarrow \infty$, the condition under which Theorem 1 applies.*

Part (b) applies to any cost charged per deviation regardless of context, such as a human review or a payment. With such costs the loss of boundary inference under a common temperature diverges too, and the contrast of Section 6 disappears. Part (b) concerns this design only; it does not compare the least possible costs of the two targets. The proof is in Appendix A.

5 Population effects require paid exploration

Fix $B > 0$ and $\sigma > 0$. Let $\mathcal{M}(B, \sigma)$ be the model in which $Y \mid A = a, H = h$ is $N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 are *any* measurable functions bounded by B . Nothing in this model links effects near the boundary to effects elsewhere.

Theorem 2 (Information–exploration inequality). *Let $S_1, \dots, S_J$ be disjoint sets with $f_j = F(S_j) > 0$, each contained in $\{a^* = 0\}$ or in $\{a^* = 1\}$, and $g_j = \text{ess inf}_{S_j} g$. For every design, adaptive or not, and every estimator $\hat{\theta}$,*

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \cdot \left(\sup_{\mathcal{M}(B, \sigma)} R_n + \frac{\sigma^2 \pi^2}{B^2} \sum_j g_j \right) \geq \sigma^2 \left(\sum_j f_j \sqrt{g_j} \right)^2. \quad (4)$$

Consequently, if $F(g \geq \gamma) > 0$ for some $\gamma > 0$, uniform consistency requires $\sup R_n \rightarrow \infty$; and if $\sup \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$, then $\liminf_n \delta_n^2 \sup R_n \geq \sigma^2 (\mathbb{E} \sqrt{g(H)})^2$.

The sample size does not appear in (4), so the lower bound on the exploration loss needed for a given precision does not decrease as the deployment grows. At a fixed

precision δ , the inequality gives $\sup R_n \geq \sigma^2(\sum_j f_j \sqrt{g_j})^2/\delta^2 - \sigma^2 \pi^2 \sum_j g_j/B^2$. The best partition gives at most $\sigma^2(\mathbb{E}\sqrt{g})^2/\delta^2$, and that constant is reached only as $\delta \rightarrow 0$. At $R_n = 0$ and $\sum_j g_j > 0$ the bound is the finite floor $B^2(\sum_j f_j \sqrt{g_j})^2/(\pi^2 \sum_j g_j)$, as it should be, since a constant estimator has bounded error.

Proposition 2 (Achievability). *Suppose $0 < \mathbb{E}\sqrt{g} < \infty$, $\delta_n > 0$ and $n\delta_n^2 \rightarrow \infty$. Write $C = \sigma^2 + B^2$. For $n\delta_n^2 > 4C$, define*

$$c_n = \frac{\mathbb{E}\sqrt{g}}{n\delta_n^2/C - 4}, \quad q_n(h) = \begin{cases} \min\{1/2, c_n/\sqrt{g(h)}\}, & g(h) > 0, \\ 1/2, & g(h) = 0, \end{cases} \quad (5)$$

and let $e_n(h)$ be the corresponding probability of action 1. The Horvitz–Thompson estimator

$$\hat{\theta}_{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \left\{ \frac{A_i Y_i}{e_n(H_i)} - \frac{(1 - A_i) Y_i}{1 - e_n(H_i)} \right\} \quad (6)$$

has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta}_{\text{HT}} - \theta)^2 \leq \delta_n^2$ and $R_n \leq Cn(\mathbb{E}\sqrt{g})^2/(n\delta_n^2 - 4C) = C(\mathbb{E}\sqrt{g})^2\delta_n^{-2}(1 + o(1))$. If also $\mathbb{E}[g] < \infty$, uniform mixing with $q \equiv C/(n\delta_n^2 - 2C)$, for $n\delta_n^2 \geq 4C$, has the same uniform risk guarantee and $R_n = Cn\mathbb{E}[g]/(n\delta_n^2 - 2C) = C\mathbb{E}[g]\delta_n^{-2}(1 + o(1))$.

The lower bound and the attainable loss share the order δ^{-2} and the functional $(\mathbb{E}\sqrt{g})^2$. Their constants are σ^2 and $\sigma^2 + B^2$, which can be far apart when $B^2 \gg \sigma^2$, and we have not found the optimal one. Proofs are in Appendix B.

6 The separation in one model

Theorem 2 is proved in $\mathcal{M}(B, \sigma)$, which places no restriction away from the boundary. A lower bound in a large model says nothing about a smaller one, so we now work with a single model that is smooth everywhere. Fix F satisfying (D1), $g = \phi(|h|)$ satisfying (D5), and $B, L, \sigma > 0$. Let $\mathcal{M}_{\text{Lip}}$ be the set of outcome laws with $Y \mid A = a, H = h \sim N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 bounded by B and L -Lipschitz on $\mathbb{R}$. Every member of $\mathcal{M}_{\text{Lip}}$ satisfies (D2) and (D3).

Assumption 2. There are $0 < u_1 < u_2$ with $F([u_1, u_2]) > 0$ and $\gamma := \text{ess inf}_{h \in [u_1, u_2]} g(h) > 0$.

Theorem 3 (Separation). *Under Assumption 2, let $b(h) = \sin^2(\pi(h - u_1)/(u_2 - u_1))\mathbf{1}\{u_1 \leq h \leq u_2\}$, $t_{\max} = \min\{B, L(u_2 - u_1)/\pi\}$, and for $|t| \leq t_{\max}$ let μ^t have $\mu_1^t \equiv 0$ and $\mu_0^t = -tb$. Then $\mu^t \in \mathcal{M}_{\text{Lip}}$, all μ^t coincide on $(-\infty, u_1]$, $\beta_0(\mu^t) = 0$ and $\theta(\mu^t) = t\mathbb{E}[b(H)]$.*

(S1) *Under the common-temperature design $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, $R_n \rightarrow 0$ and $\sup_{\mu \in \mathcal{M}_{\text{Lip}}} |P_\mu\{\beta_0(\mu) \in \hat{\beta} \pm z_{1-\eta/2}\hat{s}/\sqrt{n}\} - (1 - \eta)| \rightarrow 0$.*

(S2) *For every design and every t , $\text{TV}(P_{\mu_0^t}^n, P_{\mu^t}^n) \leq R_n(\mu^0)/\gamma$. Hence if $R_n(\mu^0) \rightarrow 0$, for every $t_1 \in (0, t_{\max}]$ and every estimator, $\max_{t \in \{0, t_1\}} P_{\mu^t}(|\hat{\theta}_n - \theta(\mu^t)| \geq t_1 \mathbb{E}[b]/2) \geq (1 - R_n(\mu^0)/\gamma)/2 \rightarrow 1/2$.*

(S3) For every design and every estimator,

$$\sup_{\mathcal{M}_{\text{Lip}}} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \frac{(\mathbb{E}[b])^2}{\sup_{\mathcal{M}_{\text{Lip}}} R_n / (\sigma^2 \gamma) + \pi^2 / t_{\max}^2}.$$

Both halves are now uniform over the same model. (S1) gives boundary intervals with uniformly correct coverage, (S3) shows that uniform consistency for the population effect needs unbounded worst-case loss, and (S2) makes the same point for two fixed members. Since $\theta \in [-2B, 2B]$, an estimator can be clipped to that interval, so consistency in probability and in mean square are equivalent here. Whether the constant of Theorem 2 holds in $\mathcal{M}_{\text{Lip}}$ is open. In submodels where boundary data identify θ the perturbation family is not available, and (S2) and (S3) do not apply. Proofs are in Appendix C.

7 Temperature as an exploration device

Theorem 4 (One common temperature). *Consider common-temperature designs with one deterministic τ_n per n . Assume $g = \phi(|\Delta|)$ with $\mathbb{E}[g] < \infty$; $|\Delta(H)|$ has a density at least $f_{\min} > 0$ on $[0, v_0]$; $\phi(u) \geq \kappa u$ on $[0, v_0]$; and $D_{\max} = \text{ess sup } |\Delta(H)| < \infty$. Fix $\epsilon \in (0, D_{\max})$ and let $\nu_\epsilon = \max_j \mathbb{P}(|\Delta(H)| \geq D_{\max} - \epsilon, a^*(H) = j)$, which is positive. If an estimator has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$ with $\delta_n^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$ and $2n\delta_n^2 > \nu_\epsilon \sigma^2$, then*

$$R_n \geq \kappa f_{\min} c_* n \min \left\{ v_0^2, \frac{(D_{\max} - \epsilon)^2}{\log^2 \{2n\delta_n^2 / (\nu_\epsilon \sigma^2)\}} \right\}, \quad c_* = \int_0^1 v \text{sig}(-v) dv.$$

The precision δ_n may be fixed. For fixed δ with $\delta^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$, a common temperature needs loss of order at least $n / \log^2 n$, while Proposition 2 reaches the same precision with loss bounded in n . Some condition on δ is needed: for $\delta \geq 2B$ the estimator that always returns zero meets the target without exploring. The theorem holds in $\mathcal{M}(B, \sigma)$; with a correctly specified parametric outcome model, a model-based estimator can do better. A common temperature explores most near the boundary, and so does the optimal design, so the location of exploration is not the issue. The issue is that the probability of the non-preferred action decays exponentially in $|\Delta|$. Confident contexts are then almost never explored unless the temperature is raised everywhere, which increases the loss incurred in contexts near the boundary.

Remark 1 (Mixtures of temperatures). Theorem 4 does not extend to random mixtures of finite temperatures. With $|\Delta| = |H|$, $H \sim U(-1, 1)$ and $g = |H|$, let $r(u) = \text{sig}(-u)$ be the non-preferred probability at temperature 1, and mix temperature 1 with weight $w_n = \mathbb{E}[1/r(|H|)] / \{n\delta^2 / (\sigma^2 + B^2) - 2\}$ and temperature $1/n$ otherwise. The Horvitz-Thompson estimator then has uniform mean squared error at most δ^2 over $\mathcal{M}(B, \sigma)$, and the loss is at most $(\sigma^2 + B^2) \mathbb{E}[1/r] \mathbb{E}[|H|r] / \delta^2 + O(1/n)$, bounded in n ; at $\delta = 0.1$ the factor $\mathbb{E}[1/r] \mathbb{E}[|H|r] / \delta^2$ equals 46.36. What is costly is one temperature shared by all contexts.

Proposition 3 (Context temperatures). *For $\Delta(h) \neq 0$ and $q(h) \in (0, 1/2)$, the context temperature $\tau(h) = |\Delta(h)| / \log\{(1 - q(h))/q(h)\}$ gives non-preferred probability $q(h)$; $q(h) = 1/2$ requires $\tau(h) = \infty$, and on $\{\Delta = 0\}$ every temperature gives $q = 1/2$. If $g = \phi(|\Delta|)$ with $\phi(0) = 0$, the design q_n of Proposition 2 is implementable with temperatures in $(0, \infty]$, and for each h with $\Delta(h) \neq 0$ and $g(h) > 0$, $\tau_n(h) \log(1/c_n) \rightarrow |\Delta(h)|$.*

The limit holds for each h . Where the cap binds, for example $0 < |h| \leq 4c_n^2$ when $g(h) = |h|$, the temperature is $\tau = \infty$. If the gap is known only up to a multiplicative error, Lemma 1 in Appendix D bounds the extra cost of allocating by the wrong gap. Proofs are in Appendix D.

8 Simulations

The simulations use $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $\mu_0(h) = h$, $c(h) = 1 + |h|$ (population effect 1.5, boundary effect 1) and $N(0, 1)$ noise. Code, seeds and result files are in the repository cited in the data availability statement.

Exploration designs. To compare allocations we hold fixed the sparse-exploration criterion $V_{\text{sp}}(q) = \sigma^2 \mathbb{E}[1/q]/n$. This is a criterion for allocation, not the mean squared error of a particular estimator. At $n = 10^5$ and $V_{\text{sp}} = 0.01$ the exploration loss is 50.0 for uniform mixing, 44.4 for the gap-based design $q \propto g^{-1/2}$ and 988.2 for a common temperature ($\tau = 0.110$), about 19.8 times the loss of uniform mixing (Figure 3; Figure 5 in Appendix E repeats this at $V_{\text{sp}} = 0.03^2$). The gap-based design here is calibrated to V_{sp} and capped at $1/2$, rather than using the c_n of Proposition 2. Table 1 runs the augmented inverse-propensity estimator on common random numbers, with a working model linear in $(1, H, |H|)$ in each arm, which is correct for this design. Implementing the gap-based design through context temperatures gives identical rows. Because V_{sp} is fixed, the expected number of non-preferred actions stays bounded for uniform mixing and the gap-based design, so the nominal coverage of their Wald intervals has no asymptotic justification as n grows. In a two-context example with masses $(0.5, 0.5)$, gaps $(0.2, 1)$, $\sigma = 1$, $B = 5$ and $\delta = 0.1$, the bound (4) equals 51.9, against an asymptotic constant of 52.4, and the common-temperature loss grows locally like n to the power 0.808, close to the predicted $1 - \Delta_{\min}/\Delta_{\max} = 0.8$.

Boundary inference. Table 2 reports the estimator of Theorem 1. The exploration loss agrees with the constant in Theorem 1(v). With $\zeta = 0.6$ coverage is close to nominal: at $n = 10^6$, 5,000 replications give 0.9458 (MCSE 0.0032), and the estimated standard error is 0.993 times the Monte Carlo standard deviation. When the boundary sample is very small the interval under-covers, with 0.887 at $n\tau_n \approx 10$. We also checked the first-order approximation used in Figure 2(b), with 4,000 replications per setting and undefined estimates set to zero. The ratio of Monte Carlo to approximate root mean squared error lies between 0.994 and 1.023 in the 16 settings with $n\tau \geq 3$. At $n\tau = 0.33$, where 0.25% of the estimates were undefined, it is 0.53.

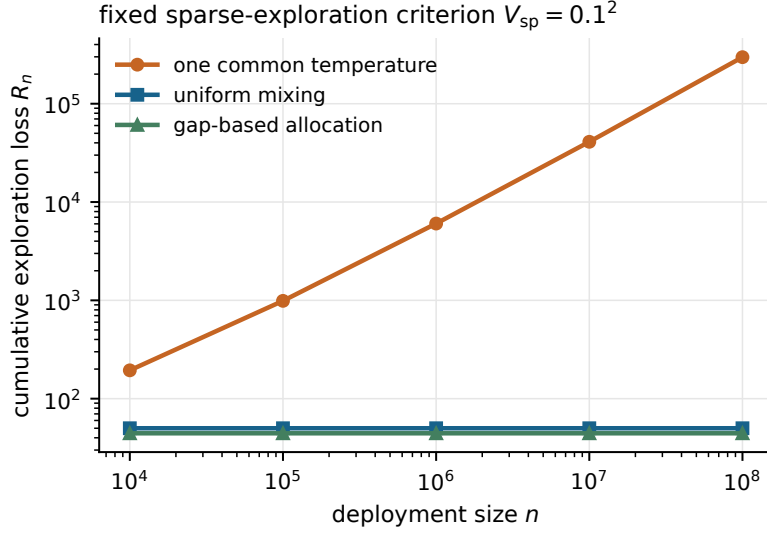


Figure 3: Exploration loss against deployment size, in closed form, with the sparse-exploration criterion fixed at $V_{\text{sp}} = 0.1^2$.

δ	design	RMSE	coverage	MCSE	explored	loss R_n
0.10	uniform mixing	0.098	0.950	0.011	100	49.9
0.10	$q \propto g^{-1/2}$ (= context temp.)	0.103	0.932	0.013	134	44.5
0.10	common temperature	0.097	0.965	0.009	7607	988.5
0.05	uniform mixing	0.052	0.955	0.010	400	200.3
0.05	$q \propto g^{-1/2}$ (= context temp.)	0.053	0.940	0.012	532	178.0
0.05	common temperature	0.050	0.948	0.011	9191	1440.1

Table 1: Exploration designs at $n = 10^5$, 400 replications with common random numbers. Coverage is of the population effect by Wald intervals; MCSE uses the observed coverage. “Explored” is the mean number of non-preferred actions.

ζ	n	$n\tau_n$	reps	RMSE	coverage of β_0	MCSE	loss R_n (Thm. 1)
0.60	10,000	39.8	1,000	0.215	0.949	0.007	1.3×10^{-1} (1.3×10^{-1})
0.60	100,000	100.0	500	0.141	0.944	0.010	8.2×10^{-2} (8.2×10^{-2})
0.60	1,000,000	251.2	200	0.082	0.970	0.012	5.1×10^{-2} (5.2×10^{-2})
0.75	10,000	10.0	1,000	0.464	0.887	0.010	8.1×10^{-3} (8.2×10^{-3})
0.75	100,000	17.8	500	0.339	0.924	0.012	2.6×10^{-3} (2.6×10^{-3})
0.75	1,000,000	31.6	200	0.247	0.930	0.018	8.2×10^{-4} (8.2×10^{-4})
0.60	1,000,000	251.2	5,000	0.090	0.946	0.003	—

Table 2: Boundary inference under common temperature $\tau_n = n^{-\zeta}$. Coverage is of $\beta_0 = c(0)$. Loss is the Monte Carlo mean with the Theorem 1(v) constant in parentheses. The last row is a separate 5,000-replication run.

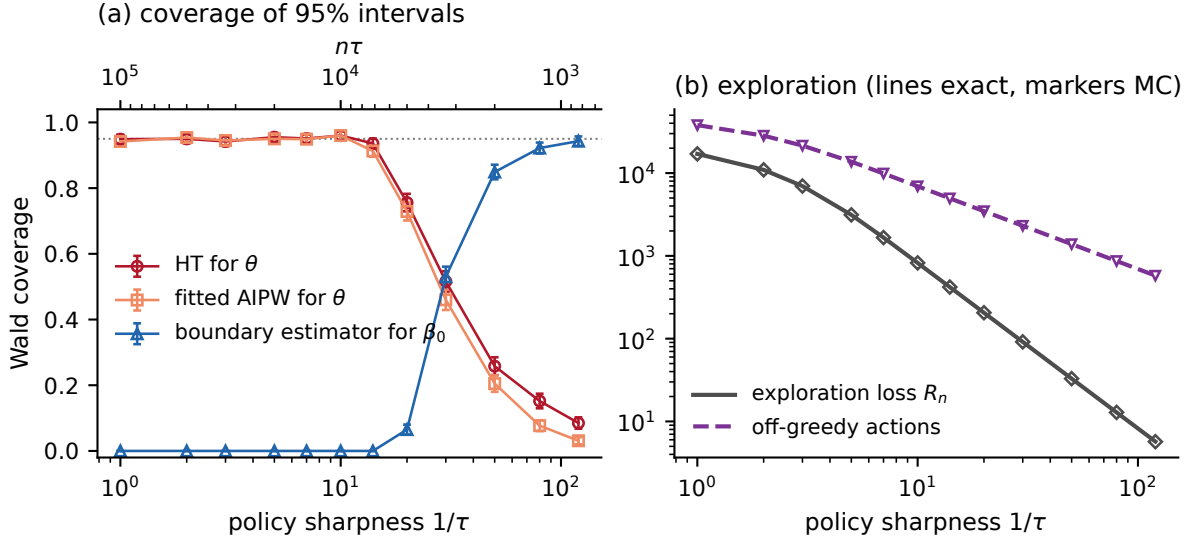


Figure 4: Sharpening a common temperature at $n = 100,000$. (a) Coverage of Wald intervals, with 95% Monte Carlo bands; the top axis shows $n\tau$. (b) Exploration loss and expected number of non-preferred actions (lines exact, markers Monte Carlo).

Temperature sweep. Figure 4 fixes $n = 100,000$ and sharpens a common temperature, with 1,000 replications per temperature. Coverage of the population effect falls from 0.949 at $1/\tau = 1$ to 0.085 for the Horvitz–Thompson estimator and 0.031 for the augmented estimator at $1/\tau = 120$. Coverage of the boundary effect rises from 0.000 to 0.943. At $1/\tau = 1$ the interval is centred on $\beta_{\text{ov},\tau}$, which is 0.48 away from β_0 , and the gap closes as τ falls. Meanwhile the exploration loss drops from 17,056 to 5.7. Over this range $n\tau$ stays above 833; pushing the temperature lower at fixed n would eventually break the boundary interval as well (Table 2). Once the policy is sharp, Monte Carlo understates the error of the Horvitz–Thompson estimator. At $1/\tau = 120$ its Monte Carlo root mean squared error is 0.36, but its exact standard deviation is of order 10^{22} , driven by rare and very large terms that the replications never draw (Figure 6 in Appendix E).

A local-linear benchmark. An analyst could also ignore the known probabilities and treat the logs as a fuzzy regression-discontinuity design at $\Delta = 0$, dividing the jump in a local-linear fit of Y by the jump in a fit of A , with bandwidth b . For $\tau > 0$ the assignment probability is continuous at zero, so this is only an approximation to such a design, and its limit is not β_0 in general. The observed mean $\mathbb{E}[Y \mid H = h] = h + e_\tau(h)(1 + |h|)$ is curved on both sides even though the potential-outcome means are linear; at $1/\tau = 30$ and $b = 0.2$ the estimator converges to 0.935 instead of 1. Table 3 in Appendix E compares the two estimators on the same logs at $n = 10^5$. Neither works well at $1/\tau \leq 30$. At $1/\tau = 120$ both cover at about the nominal rate and the local-linear estimator with $b = 0.2$ is more accurate, and at $1/\tau = 1000$ it is more accurate at both bandwidths. These findings are specific to this design. They bear on the choice of boundary estimator, whereas the lower

bounds for the population effect apply to every estimator.

9 Discussion

For the boundary effect, when deviations near the boundary have small operational cost, a growing deployment under one sharpening temperature gives increasingly precise inference while the cumulative exploration loss tends to zero. For the population average effect, uniformly consistent estimation requires the worst-case exploration loss to diverge, and increasing the deployment size does not remove this lower bound on the required loss. A report of an effect estimated from such logs should therefore state which of the two quantities it targets.

For design, the question is how to use temperature. With one temperature for all contexts, the probability of the non-preferred action falls off exponentially with the policy’s confidence, so reaching confident contexts requires a higher temperature, which increases the loss incurred near the boundary. Uniform mixing, context-specific temperatures and mixtures that put a vanishing weight on a moderate temperature all avoid this. Allocating by the gap saves a little more when gaps vary; in the example of Section 8 the saving over uniform mixing is about eleven percent.

Limitations. The policy is known and its assignments are logged, and the score does not learn from outcomes. Each unit receives one decision and contexts are independent. The boundary results assume a one-dimensional score. The lower bounds assume Gaussian outcomes with known variance and exclude models in which boundary data identify the population effect, and the separation needs a cost that vanishes at the boundary. The treatment is the choice of strategy under a fixed generator, not the text that is produced. We leave learned scores, multidimensional scores (where a natural target is the effect given $\Delta(H) = 0$) and multi-turn conversations for future work.

Data availability statement

No new participant data were collected or analysed. The results are theoretical, deterministic numerical illustrations, or simulations. The simulation code, the result files (with arguments, random seeds and code commits), plain-text versions of the proofs, and the scripts that generate every number, table and figure in the paper are available in the repository

<https://github.com/kunhailP/exploration-cost-boundary-population>

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The author confirms the sole responsibility for the conception of the study, presented results and manuscript preparation.

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The author states no conflict of interest.

Ethical approval

The conducted research is not related to either human or animals use.

Use of AI-assisted tools

Generative AI tools were used solely for minor manuscript editing, such as syntax checking and typo correction, and their application in coding was strictly limited to routine auto-completion. The author takes full responsibility for the final manuscript.

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A Proofs for Section 4

Write $e = e_\tau(H) = \text{sig}(H/\tau) = \mathbb{P}(A = 1 \mid H)$ and $\omega = e(1 - e) = \Omega(H/\tau)$ with $\Omega(u) = \exp(u)/\{1 + \exp(u)\}^2$, and $m_\tau = \mathbb{E}\omega$. We use

$$\begin{aligned} \int \Omega = 1, \quad \int |u|\Omega(u) du &= 2 \log 2, \quad \int \Omega(u) \text{sig}(u) du = \frac{1}{2}, \\ \Omega(u) &\leq \exp(-|u|), \quad \int_0^\infty \text{sig}(-v) dv = \log 2, \quad \int_0^\infty v \text{sig}(-v) dv = \frac{\pi^2}{12}. \end{aligned}$$

A.1 Proof of Theorem 1

Step 1 (kernel integrals). For ϱ bounded and continuous at 0, $j \geq 1$ and $r, s \geq 0$, substituting $h = \tau u$ and using dominated convergence with dominating function $\|\varrho\|_\infty \bar{f}\Omega(u)$,

$$\mathbb{E}[\Omega(H/\tau)^j (1 - e)^r e^s \varrho(H)] = \tau f(0) \varrho(0) \int \Omega^j \text{sig}(-u)^r \text{sig}(u)^s du + o(\tau).$$

In particular $m_\tau = \tau f(0)(1 + o(1))$.

Step 2 (bias). $\beta_{\text{ov},\tau} - \beta_0 = \mathbb{E}[\omega\{c(H) - c(0)\}]/m_\tau$. On $|H| \leq u_0$, $|c(h) - c(0)| \leq 2L|h|$ and $\mathbb{E}[\omega|H|] \leq \bar{f}\tau^2 \int |u|\Omega = 2\log 2 \bar{f}\tau^2$. On $|H| > u_0$, $\omega \leq \exp(-u_0/\tau)$ and $|c(h) - c(0)| \leq 4B$. Hence $|\beta_{\text{ov},\tau} - \beta_0| \leq (4\log 2 L \bar{f}\tau^2 + 4B \exp(-u_0/\tau))/m_\tau = (4\log 2 L \bar{f}/f(0))\tau(1+o(1))$.

Step 3 (linearization). Since $\mathbb{E}[w_1 | H] = \omega$ and $\mathbb{E}[w_1 Y | H] = \omega\mu_1$, $\mathbb{E}[w_1(Y - \bar{\mu}_1)] = 0$, and likewise for arm 0. Define $T_a = n^{-1} \sum_i w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$, so that $\hat{m}_a - \bar{\mu}_a = (m_\tau/\hat{D}_a)T_a$, $T_1 - T_0 = n^{-1} \sum_i \xi_i$ and $\mathbb{E}\xi = 0$. Because $w_1 w_0 = 0$ and $\mathbb{E}[A(1-e)^2 | H] = \omega(1-e)$,

$$s_n^2 = m_\tau^{-2} \mathbb{E}[\omega\{(1-e)(\sigma_1^2 + (\mu_1 - \bar{\mu}_1)^2) + e(\sigma_0^2 + (\mu_0 - \bar{\mu}_0)^2)\}]. \quad (7)$$

By Step 1, $\bar{\mu}_a \rightarrow \mu_a(0)$; the terms in $(\mu_a - \bar{\mu}_a)^2$ are $o(\tau)$ by dominated convergence; and the variance terms equal $\tau f(0)\sigma_a^2(0)/2 + o(\tau)$. Hence $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau f(0)\}(1+o(1))$.

Step 4 (central limit theorem). Since $0 \leq w_a \leq 1$, $w_a^4 \leq w_a$, and $\mathbb{E}[(Y - \bar{\mu}_a)^4 | A, H] \leq C(M_4, B)$, we get $\mathbb{E}[w_a^4(Y - \bar{\mu}_a)^4] \leq C m_\tau$ and $\mathbb{E}\xi^4 \leq C' m_\tau^{-3}$. The Lyapunov ratio $n^{-1}\mathbb{E}\xi^4/s_n^4 = O((n\tau)^{-1}) \rightarrow 0$, so $n^{-1/2} \sum \xi_i/s_n \Rightarrow N(0, 1)$. Also $\text{Var}(\hat{D}_a)/m_\tau^2 \leq 1/(nm_\tau) \rightarrow 0$, so $\hat{D}_a/m_\tau \rightarrow 1$ in probability and $m_\tau/\hat{D}_a - 1 = O_p((n\tau)^{-1/2})$. Now

$$\hat{\beta} - \beta_{\text{ov},\tau} = (T_1 - T_0) + (m_\tau/\hat{D}_1 - 1)T_1 - (m_\tau/\hat{D}_0 - 1)T_0, \quad (8)$$

with $T_a = O_p((n\tau)^{-1/2})$. The remainder is $O_p((n\tau)^{-1}) = o_p(s_n/\sqrt{n})$, and Slutsky's lemma gives (i). The event that a weight sum is zero has probability at most $2(1 - m_\tau)^n \leq 2\exp(-nm_\tau) \rightarrow 0$, because $\mathbb{P}(w_a > 0) \geq \mathbb{E}w_a = m_\tau$.

Step 5 (variance estimator). Write $\xi_{ai} = w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$. First, $n^{-1} \sum \xi_i^2/s_n^2$ has mean one and variance at most $\mathbb{E}\xi^4/(ns_n^4) = O((n\tau)^{-1})$. Second,

$$\hat{\psi}_{1i} - \xi_{1i} = w_{1i}(Y_i - \bar{\mu}_1)(1/\hat{D}_1 - 1/m_\tau) + w_{1i}(\bar{\mu}_1 - \hat{m}_1)/\hat{D}_1.$$

The mean square of the first term is $(n^{-1} \sum \xi_{1i}^2)(m_\tau/\hat{D}_1 - 1)^2 = O_p(s_n^2)O_p((n\tau)^{-1})$. For the second, $w_1^2 \leq w_1$ gives mean square at most $(\hat{m}_1 - \bar{\mu}_1)^2/\hat{D}_1$; by Step 4 and $\hat{D}_1 = m_\tau(1 + o_p(1))$ this is $O_p(1/(nm_\tau^2))$, which is $O_p((n\tau)^{-1})$ relative to $s_n^2 \asymp \tau^{-1}$. Arm 0 is identical. By Minkowski's and the Cauchy-Schwarz inequalities, $|n^{-1} \sum \hat{\psi}_i^2 - n^{-1} \sum \xi_i^2| = o_p(s_n^2)$, which gives (iii).

Step 6 (centering at β_0). $(\beta_{\text{ov},\tau} - \beta_0)/(s_n/\sqrt{n}) = O(\tau)O((n\tau)^{1/2}) = O((n\tau^3)^{1/2})$. With (i), (iii) and Slutsky's lemma this gives (iv).

Step 7 (exploration loss). The non-preferred probability is $q_\tau(h) = \text{sig}(-|h|/\tau)$. On $|h| \leq u_0$, $n\mathbb{E}[\phi(|H|)\text{sig}(-|H|/\tau)] \leq n\bar{\kappa}\bar{f} \cdot 2\tau^2 \int_0^\infty v\text{sig}(-v)dv = n\bar{\kappa}\bar{f}\tau^2\pi^2/6$, and the rest is at most $n\bar{G}\exp(-u_0/\tau)$. In the strong form, on $v \leq u_0/\tau$, $\phi(\tau v)/\tau \rightarrow \kappa v$ and $f(\tau v) \rightarrow f(0)$ with dominating function $\bar{\kappa}v\bar{f}\text{sig}(-v)$, which gives the asymptotic equality. For $\tau_n = n^{-\zeta}$: $n\tau_n \rightarrow \infty$ iff $\zeta < 1$; $n\tau_n^3 \rightarrow 0$ iff $\zeta > 1/3$; $n\tau_n^2 \rightarrow 0$ iff $\zeta > 1/2$; and $n\exp(-u_0n^\zeta) \rightarrow 0$. The expected number of non-preferred actions is $n\mathbb{E}[q_\tau(H)] = n\tau \int f(\tau u)\text{sig}(-|u|)du = 2\log 2 f(0) n\tau(1 + o(1))$ by the same dominated convergence argument. $\square$

Smoother effects. If $|c(h) - c(0) - c'(0)h| \leq M_2 h^2$ and f is L_f -Lipschitz near 0, then $\mathbb{E}[\omega H] = \tau^2 \int u \Omega(u) \{f(\tau u) - f(0)\} du = O(\tau^3)$ up to an exponentially small tail, and the quadratic remainder contributes at most $M_2 \bar{f} \tau^3 \int u^2 \Omega(u) du / m_\tau$; the bias is $O(\tau^2)$. Differentiability of c alone is not enough: a bounded effect equal to $c(h) = 1 + |h|^{3/2}$ near zero gives bias $\tau^{3/2} \int |u|^{3/2} \Omega(u) du (1 + o(1))$.

A.2 Proof of Proposition 1

(a) Substituting $h = \tau v$ on $|h| \leq u_0$ and using $\text{sig}(-v) \leq \exp(-v)$ elsewhere,

$$R_n \leq n \bar{\kappa} \bar{f} \int_{|h| \leq u_0} |h|^p \text{sig}(-|h|/\tau) dh + n \bar{G} \exp(-u_0/\tau) \leq 2 \bar{\kappa} \bar{f} C_p n \tau^{p+1} + n \bar{G} \exp(-u_0/\tau).$$

Here $C_p = \Gamma(p+1)(1 - 2^{-p})\zeta_R(p+1)$ with ζ_R the Riemann zeta function, which follows from expanding $\text{sig}(-v) = \sum_{k \geq 1} (-1)^{k+1} \exp(-kv)$; for example $C_{1/2} \approx 0.678$, $C_1 = \pi^2/12$ and $C_2 \approx 1.803$. Validity of the interval uses only (D1)–(D4) and $1/3 < \zeta < 1$, by Theorem 1(iv), and $n \tau_n^{p+1} \rightarrow 0$ iff $\zeta > 1/(p+1)$.

(b) Restricting to $0 < |H| \leq u_0$ and substituting $h = \tau u$,

$$R_n \geq n \lambda_0 \tau \int_{|u| \leq u_0/\tau} f(\tau u) \text{sig}(-|u|) du = 2 \log 2 \lambda_0 f(0) n \tau (1 + o(1)),$$

by dominated convergence with dominating function $\bar{f} \text{sig}(-|u|)$. $\square$

B Proofs for Section 5

Proof of Theorem 2. *Submodel.* For $\vartheta \in [-B, B]^J$ set $\mu_{a^*(h)}^\vartheta(h) = 0$, and on S_j set $\mu_{\bar{a}(h)}^\vartheta(h) = s_j \vartheta_j$ with $s_j = +1$ if $\bar{a} = 1$ on S_j and $s_j = -1$ otherwise; elsewhere $\mu_{\bar{a}(h)}^\vartheta(h) = 0$. Then $\mu^\vartheta \in \mathcal{M}(B, \sigma)$ and $\theta(\mu^\vartheta) = \sum_j f_j \vartheta_j$.

Information. The sample density is $\prod_i dF(H_i) \pi_i(A_i | H_i, \mathcal{F}_{i-1}) \varphi_\sigma(Y_i - \mu_{A_i}^\vartheta(H_i))$, where the assignment kernels π_i are known functions of observed data and do not depend on ϑ . The score for ϑ_j is $\sum_i \mathbf{1}\{H_i \in S_j, A_i = \bar{a}(H_i)\} s_j (Y_i - s_j \vartheta_j) / \sigma^2$. Its summands are martingale differences, so the Fisher information matrix is diagonal with $I_j(\vartheta) = \mathbb{E}_\vartheta N_j / \sigma^2$, where $N_j = \sum_i \mathbf{1}\{H_i \in S_j, A_i = \bar{a}(H_i)\}$.

van Trees. Take the product prior with marginal density $B^{-1} \cos^2(\pi x / (2B))$ on $[-B, B]$, whose Fisher information is $J_0 = \pi^2 / B^2$. With $\bar{N}_j = \int \mathbb{E}_\vartheta N_j \lambda(d\vartheta)$, the multi-variate van Trees inequality [43], maximized over the direction vector, gives

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \sum_j \frac{f_j^2}{\bar{N}_j / \sigma^2 + J_0} = \sum_j \frac{f_j \sigma^2}{a_j}, \quad a_j = \frac{\bar{N}_j + \sigma^2 J_0}{f_j}.$$

Cauchy–Schwarz and loss. Writing $\sigma \sum_j f_j \sqrt{g_j} = \sum_j \sqrt{f_j g_j a_j} \cdot \sigma \sqrt{f_j / a_j}$,

$$\sigma^2 \left(\sum_j f_j \sqrt{g_j} \right)^2 \leq \left(\sum_j f_j g_j a_j \right) \left(\sum_j f_j \sigma^2 / a_j \right).$$

Here $\sum_j f_j g_j a_j = \sum_j g_j \bar{N}_j + \sigma^2 J_0 \sum_j g_j$, and since $g \geq g_j$ almost surely on S_j , $\sum_j g_j N_j \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\sum_j g_j \bar{N}_j \leq \sup R_n$. This proves (4).

Consequences. With $J = 1$ and $S_1 = \{g \geq \gamma\} \cap \{a^* = j\}$ of positive mass for some j , (4) gives $\sup R_n \geq g_1 \{\sigma^2 f_1^2 / \sup \mathbb{E}(\hat{\theta} - \theta)^2 - \sigma^2 \pi^2 / B^2\} \rightarrow \infty$. For the constant, fix a partition into level sets of g intersected with the two preferred-action sides; (4) gives $\liminf \delta_n^2 \sup R_n \geq \sigma^2 (\sum f_j \sqrt{g_j})^2$, and the lower sums $\sum F(S) \text{ess inf}_S \sqrt{g}$ increase to $\mathbb{E} \sqrt{g}$ as the level mesh shrinks. $\square$

Proof of Proposition 2. The Horvitz–Thompson estimator is unbiased with variance at most $n^{-1} C \mathbb{E}[1/e + 1/(1 - e)] = n^{-1} C \mathbb{E}[1/q + 1/(1 - q)] \leq n^{-1} C (\mathbb{E}[1/q] + 2)$, since $\{e, 1 - e\} = \{q, 1 - q\}$ and $q \leq 1/2$. Pointwise $1/q = \max\{2, \sqrt{g}/c_n\} \leq \sqrt{g}/c_n + 2$, and the choice of c_n gives $\mathbb{E}[1/q] + 2 \leq n \delta_n^2 / C$. The loss is $n \mathbb{E}[gq] \leq n c_n \mathbb{E} \sqrt{g}$. For uniform mixing with $q = C/(n \delta_n^2 - 2C) \leq 1/2$, the same variance bound gives $C(1/q + 2)/n = \delta_n^2$, and the loss is $nq \mathbb{E}[g] = C \mathbb{E}[g] n / (n \delta_n^2 - 2C)$. Without the correction, $q = C/(n \delta_n^2)$ gives risk $C/\{nq(1 - q)\} > \delta_n^2$ at $\mu_0 = \mu_1 \equiv B$, where the bound is attained. $\square$

C Proof of Theorem 3

The function b satisfies $0 \leq b \leq 1$, is $\pi/(u_2 - u_1)$ -Lipschitz on $\mathbb{R}$, vanishes outside $[u_1, u_2]$, and has $\mathbb{E}b(H) > 0$ because H has a density and $F([u_1, u_2]) > 0$. Hence $|tb| \leq B$ and tb is L -Lipschitz for $|t| \leq t_{\max}$, so $\mu^t \in \mathcal{M}_{\text{Lip}}$; $c^t = tb$, $\beta_0(\mu^t) = 0$, and $\theta(\mu^t) = t \mathbb{E}b(H)$. Every observation affected by the perturbation has $A = 0$ and $H \in [u_1, u_2] \subset (0, \infty)$, where $a^* = 1$, so it is a non-preferred action. Let $N = \sum_i \mathbf{1}\{A_i = 0, H_i \in [u_1, u_2]\}$.

(S1): uniform boundary coverage. The loss statement is Theorem 1(v), since the design does not depend on μ . For coverage we show that the error terms in the proof of Theorem 1 are bounded by constants that do not depend on $\mu \in \mathcal{M}_{\text{Lip}}$. Here $\sigma_a^2 \equiv \sigma^2$, and m_τ , $\hat{D}_a$ and the event of a zero weight sum depend only on (H_i, A_i) , whose law does not depend on μ . Throughout, C denotes a constant depending only on (B, L, σ, f) .

Variance bounds. Since $(1 - e) + e = 1$ and $|\mu_a - \bar{\mu}_a| \leq 2B$, (7) gives

$$\sigma^2/m_\tau \leq s_n^2 \leq (\sigma^2 + 4B^2)/m_\tau.$$

Berry–Esseen. Given (A, H) , $Y - \bar{\mu}_a$ is Gaussian with variance σ^2 and mean bounded by $2B$, so $\mathbb{E}[|Y - \bar{\mu}_a|^3 \mid A, H] \leq C$ and, as $w_a^3 \leq w_a$ and $\mathbb{E}w_a = m_\tau$, $\mathbb{E}|\xi|^3 \leq C m_\tau^{-2}$. The Berry–Esseen inequality for independent and identically distributed summands gives

$$\sup_x \left| \mathbb{P}_\mu \left(\frac{\sum_i \xi_i}{\sqrt{n} s_n} \leq x \right) - \Phi(x) \right| \leq \frac{C \mathbb{E}|\xi|^3}{\sqrt{n} s_n^3} \leq \frac{C}{\sigma^3 \sqrt{n m_\tau}},$$

uniformly in μ .

Remainder. In (8), $\text{Var}(T_a) \leq \mathbb{E}[w_a^2(Y - \bar{\mu}_a)^2]/(n m_\tau^2) \leq (\sigma^2 + 4B^2)/(n m_\tau)$, and $\text{Var}(\hat{D}_a/m_\tau) \leq 1/(n m_\tau)$ does not depend on μ . By Chebyshev's inequality, for every $\varepsilon > 0$ there is M with $\sup_\mu \mathbb{P}_\mu(|T_a| > M(n m_\tau)^{-1/2}) < \varepsilon$ and $\mathbb{P}(|m_\tau/\hat{D}_a - 1| > M(n m_\tau)^{-1/2}) < \varepsilon$.

ε for large n . Dividing by $s_n/\sqrt{n} \geq \sigma(nm_\tau)^{-1/2}$, the remainder in units of $s_n/\sqrt{n}$ is $O_p((nm_\tau)^{-1/2})$ uniformly in μ .

Variance estimator. Each bound in Step 5 of the proof of Theorem 1 involves only $\mathbb{E}\xi^4 \leq Cm_\tau^{-3}$, $s_n^2 \geq \sigma^2/m_\tau$, the bounds on T_a above and the law of $\hat{D}_a$; hence $\sup_\mu \mathbb{P}_\mu(|\hat{s}^2/s_n^2 - 1| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Bias. Global Lipschitz continuity gives, with no tail term, $|\beta_{\text{ov},\tau} - \beta_0| \leq 2L\mathbb{E}[\omega|H|]/m_\tau \leq 4\log 2 L\bar{f}\tau^2/m_\tau$, so in units of $s_n/\sqrt{n}$ the bias is at most $C\tau^2\sqrt{n/m_\tau} = O((n\tau^3)^{1/2})$ uniformly in μ , using $m_\tau = \tau f(0)(1 + o(1))$.

Coverage. Write $Z_n = (\hat{\beta} - \beta_{\text{ov},\tau})\sqrt{n}/s_n$ and $b_n = (\beta_{\text{ov},\tau} - \beta_0)\sqrt{n}/s_n$. The interval covers β_0 iff $|Z_n + b_n| \leq z_{1-\eta/2}\hat{s}/s_n$. By the three displays above, for every $\varepsilon > 0$ there are events of $\mathbb{P}_\mu$ -probability at least $1 - \varepsilon$ for all μ and large n , on which $|Z_n - n^{-1/2} \sum_i \xi_i/s_n| \leq \varepsilon$, $|b_n| \leq \varepsilon$ and $|\hat{s}/s_n - 1| \leq \varepsilon$. Hence the coverage probability lies within $\varepsilon + C/\sqrt{nm_\tau} + \sup_{|x| \leq z + 3\varepsilon} \varphi(x) \cdot 6\varepsilon$ of $1 - \eta$ for all $\mu \in \mathcal{M}_{\text{Lip}}$, where φ is the standard normal density, and letting $\varepsilon \downarrow 0$ proves (S1).

(S2). The sample densities under μ^0 and μ^t coincide on $E = \{N = 0\}$, so $P_0(E) = P_t(E)$, and for any event A , $|P_0(A) - P_t(A)| = |P_0(A \cap E^c) - P_t(A \cap E^c)| \leq \max\{P_0(E^c), P_t(E^c)\} = P_0(E^c)$. Thus $\text{TV} \leq P_0(N \geq 1) \leq \mathbb{E}_0 N$. Because $g \geq \gamma$ F -almost surely on $[u_1, u_2]$ and $H_i \sim F$, $\gamma N \leq \sum_i g(H_i)\mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\mathbb{E}_0 N \leq R_n(\mu^0)/\gamma$. Le Cam's two-point argument with separation $t_1\mathbb{E}b$ gives the displayed bound.

(S3). The score of $t \mapsto \mu^t$ is $\sum_i \mathbf{1}\{A_i = 0\}b(H_i)\{-(Y_i + tb(H_i))\}/\sigma^2$, with information $I(t) = \mathbb{E}_t \sum_i \mathbf{1}\{A_i = 0\}b(H_i)^2/\sigma^2 \leq \mathbb{E}_t N/\sigma^2 \leq R_n(\mu^t)/(\gamma\sigma^2)$. With the prior $t_{\max}^{-1} \cos^2(\pi t/(2t_{\max}))$ on $[-t_{\max}, t_{\max}]$, whose information is $\pi^2/t_{\max}^2$, the van Trees inequality for $\psi(t) = t\mathbb{E}b$ gives the bound. $\square$

D Proofs and a lemma for Section 7

Proof of Theorem 4. Take the single cell $S = \{|\Delta| \geq D_{\max} - \epsilon, a^* = j\}$, with j attaining ν_ϵ . There $q_\tau \leq \exp(-(D_{\max} - \epsilon)/\tau)$, so $\bar{N} \leq n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau)$, and the van Trees step of Theorem 2 with $J = 1$ gives $\delta_n^2 \geq \nu_\epsilon^2 \sigma^2 / (n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau) + \sigma^2 \pi^2 / B^2)$. Once $\delta_n^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$, this forces $n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau) \geq \nu_\epsilon^2 \sigma^2 / (2\delta_n^2)$, that is, $\tau_n \geq (D_{\max} - \epsilon) / \log\{2n\delta_n^2 / (\nu_\epsilon \sigma^2)\}$, where the logarithm is positive because $2n\delta_n^2 > \nu_\epsilon \sigma^2$. Nothing here requires $\delta_n \rightarrow 0$. On the other hand

$$R_n = n\mathbb{E}[\phi(|\Delta|)\text{sig}(-|\Delta|/\tau)] \geq n\underline{\kappa}f_{\min}\tau^2 \int_0^{v_0/\tau} v\text{sig}(-v) dv \geq n\underline{\kappa}f_{\min}c_* \min\{\tau, v_0\}^2.$$

Combining the two bounds proves the claim. With finitely many score values, the same argument gives $\tau \geq \Delta_{\max}/\log(C'n\delta^2)$ for δ small enough, and the $\Delta_{\min}$ cell, with mass f , contributes at least $nf\phi(\Delta_{\min})\exp(-\Delta_{\min}/\tau)/2$, which is of order $n^{1-\Delta_{\min}/\Delta_{\max}}$ up to logarithmic factors. $\square$

Remark 1. Write $u = |H|$, $C = \sigma^2 + B^2$, $r(u) = \text{sig}(-u)$, and $q_n = w_n r + (1 - w_n) \text{sig}(-nu) \leq 1/2$ for the mixture. Since $q_n \geq w_n r$, $\mathbb{E}[1/q_n] \leq \mathbb{E}[1/r]/w_n = n\delta^2/C - 2$, and the variance bound in the proof of Proposition 2 gives uniform risk at most $C(\mathbb{E}[1/q_n] + 2)/n \leq \delta^2$. The loss is $n\mathbb{E}[uq_n] \leq nw_n\mathbb{E}[ur] + n\mathbb{E}[u\text{sig}(-nu)] = C\mathbb{E}[1/r]\mathbb{E}[ur]n/(n\delta^2 - 2C) + O(1/n)$, which equals $C\mathbb{E}[1/r]\mathbb{E}[ur]/\delta^2 + O(1/n)$.

Proof of Proposition 3. Solving $\text{sig}(-|\Delta|/\tau) = q$ gives the formula. If $g(h) = 0$, then $q_n(h) = 1/2$, which is implementable by $\tau = \infty$ (and by any τ when $\Delta(h) = 0$); if $\Delta(h) \neq 0$ and the cap binds, $q_n(h) = 1/2$ again requires $\tau = \infty$. If $\Delta(h) \neq 0$ and $g(h) > 0$, then $q_n(h) = c_n/\sqrt{g(h)} < 1/2$ once $c_n < \sqrt{g(h)}/2$, and $\log\{(1 - q_n)/q_n\} = \log(1/c_n) + \frac{1}{2}\log g(h) + \log(1 - q_n)$ with $c_n \rightarrow 0$, which gives the limit. $\square$

Misspecified gaps. Suppose the allocation uses $\hat{g}$ in place of g . Designs are compared at a common value of the sparse-exploration criterion $\sigma^2\mathbb{E}[1/q]/n$, which is an allocation criterion, not the mean squared error of an estimator. Let $\Lambda(\rho) = (1 + \sqrt{\rho})^2/(4\sqrt{\rho})$.

Lemma 1. Assume $0 < \mathbb{E}\sqrt{g} < \infty$ and $0 < a \leq \hat{g}/g \leq b < \infty$ on $\{g > 0\}$, with $\hat{g} = 0$ on $\{g = 0\}$, and let $\rho = b/a$.

- (a) If $F(g > 0) = 1$ and the uncapped designs $q = c/\sqrt{\hat{g}}$ and $q' = c'/\sqrt{g}$, calibrated to the same criterion, are at most $1/2$ almost surely, the ratio of their losses is at most $\Lambda(\rho)$. The bound is attained when the measure $w \propto \sqrt{g} dF$ has a set of mass exactly one half.
- (b) For the capped designs $q_c = \min\{1/2, c/\sqrt{\hat{g}}\}$ and $q'_c = \min\{1/2, c'/\sqrt{g}\}$, equal to $1/2$ where the gap is zero and calibrated to a common value $T = \mathbb{E}[1/q_c] = \mathbb{E}[1/q'_c]$, the loss ratio converges, as $T \rightarrow \infty$, to $\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/(\mathbb{E}\sqrt{g})^2 \leq \Lambda(\rho)$.

If $\text{ess inf } g = 0$, the uncapped design exceeds $1/2$ on a set of positive mass for every $c > 0$, so part (a) is vacuous and part (b) is the relevant statement; this includes $g(h) = |h|$. A common rescaling of $\hat{g}$ changes nothing. Gap-based allocation beats uniform mixing for every admissible $\hat{g}$ if $\mathbb{E}[g]/(\mathbb{E}\sqrt{g})^2 > \Lambda(\rho)$.

Proof of Lemma 1. (a) With $q = c/\sqrt{\hat{g}}$ and $\mathbb{E}[1/q] = T$, $c = \mathbb{E}\sqrt{\hat{g}}/T$ and the loss is $n\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/T$, against $n(\mathbb{E}\sqrt{g})^2/T$ for $\hat{g} = g$. With $s = (\hat{g}/g)^{1/2} \in [\sqrt{a}, \sqrt{b}]$ and the probability measure $w \propto \sqrt{g} dF$, the ratio is $\mathbb{E}_w[s]\mathbb{E}_w[1/s]$, which Kantorovich's inequality bounds by $(\sqrt{a} + \sqrt{b})^2/(4\sqrt{ab}) = \Lambda(b/a)$. If w has a set of mass one half, setting $s = \sqrt{a}$ there and $s = \sqrt{b}$ elsewhere attains the bound. With atoms the bound need not be attained: for $F = (0.99, 0.01)$, $g = (1, 100)$ and $\rho = 36$, uniform mixing costs 1.675 times the optimum, below $\Lambda(36) = 2.042$, yet every admissible $\hat{g}$ costs at most 1.347 times the optimum.

(b) By dominated convergence, $c\mathbb{E}[1/q_c] = \mathbb{E}\max\{2c, \sqrt{\hat{g}}\} \rightarrow \mathbb{E}\sqrt{\hat{g}}$ and $\mathbb{E}[gq_c]/c \rightarrow \mathbb{E}[g/\sqrt{\hat{g}}]$, with $g/\sqrt{\hat{g}} = 0$ where $g = 0$, as $c \rightarrow 0$; likewise for q' . Calibration to a common $T \rightarrow \infty$ forces $c, c' \rightarrow 0$ with $cT \rightarrow \mathbb{E}\sqrt{\hat{g}}$ and $c'T \rightarrow \mathbb{E}\sqrt{g}$, so the loss ratio $\mathbb{E}[gq_c]/\mathbb{E}[gq'_c]$ converges to the stated limit, which is bounded as in (a). $\square$

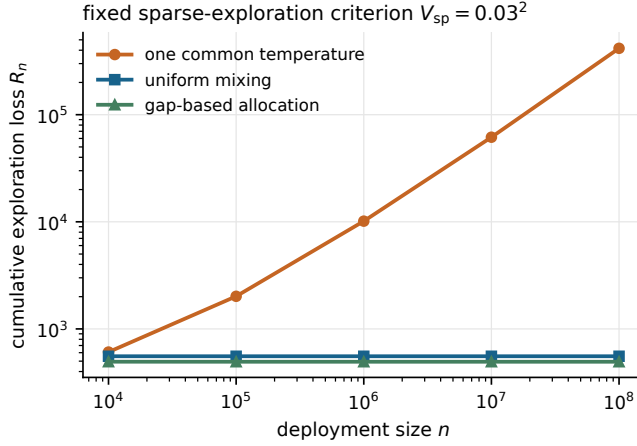


Figure 5: As Figure 3, with the sparse-exploration criterion fixed at $V_{sp} = 0.03^2$.

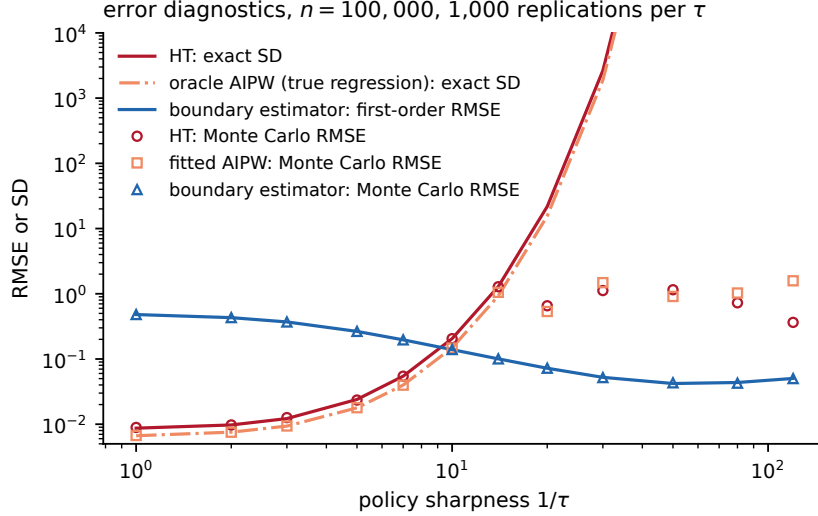


Figure 6: Error diagnostics for the runs of Figure 4. Lines: exact standard deviation of the Horvitz–Thompson estimator and of an oracle augmented estimator that uses the true regression (a reference, not an estimator used in the simulation), both for θ , and the first-order approximation to the root mean squared error of the boundary estimator of β_0 . Markers: Monte Carlo root mean squared error of the Horvitz–Thompson estimator, the augmented estimator with the fitted working model, and the boundary estimator.

E Additional figures and tables

$1/\tau$	b	Theorem 1 estimator			local-linear comparison at $\Delta = 0$			
		limit	RMSE	coverage	first stage	limit	RMSE	coverage
10	0.05	1.139	0.139	0.000	0.001	-1.400	164.8	1.000
10	0.20	1.139	0.140	0.000	0.026	0.401	34.6	0.996
30	0.05	1.046	0.052	0.536	0.012	0.733	252.6	1.000
30	0.20	1.046	0.051	0.545	0.237	0.935	0.153	0.915
120	0.05	1.012	0.050	0.945	0.237	0.984	0.261	0.962
120	0.20	1.012	0.050	0.947	0.702	0.996	0.044	0.940
1,000	0.05	1.001	0.141	0.951	0.845	1.000	0.073	0.953
1,000	0.20	1.001	0.138	0.956	0.959	1.000	0.032	0.957

Table 3: Boundary effect $\beta_0 = 1$ at $n = 10^5$, 1,000 replications, same logs for both estimators: the estimator of Theorem 1 (known probabilities) and a fuzzy local-linear comparison at $\Delta = 0$ (triangular kernel, bandwidth b , heteroskedasticity-robust delta-method standard errors). “Limit” is the probability limit, $\beta_{\text{ov},\tau}$ for the first estimator and the ratio of population local-linear gaps for the second, computed by quadrature; “first stage” is the population gap in the assignment probability. Very large errors reflect a small first stage.